A Comprehensive Introduction *to* Applicable Mathematics

Authors and contributors

Georgii Brovkin

Contents

1 Twists, knots, and other animals **1**
1.1 A hole diameter maximisation under origami twists 1

Chapter 1

Twists, knots, and other animals

1.1 A hole diameter maximisation under origami foldings (and, perhaps, small bendings)

Suppose you have a surface $\mathcal{S}$ in $\mathbb{R}^3$ (if need be it's all pretty generalisable) with a hole in it defined by its boundary, that is a simple closed path $\partial : [0,1] \to \mathbb{R}^2$ (so it is seen that the surface may have volume and the hole may not be 'flat'). Define $\mathfrak{o} : \mathcal{S} \to \mathcal{S}$ to be an "origami", that is the "twisting function". And let's define what "diameter" means: d_δ is said to be the diameter of the hole ∂ if we project orthogonally our configuration $(\mathcal{S}, \partial)$ to $(\mathcal{S}', \partial')$ respectively, where $\mathcal{S}'$ is a surface "locally parallel to the boundary ∂" (if possible); then let $d_\partial = \sup\limits_{a,b \in \mathrm{Im}(\partial)} \|f(a) - f(b)\|$.

1. Given a configuration $(\mathcal{S}, \partial)$ what is the maximum d_∂ achiavable and which origami $\mathfrak{o}$ leads to it?

2. Given a surface $\mathcal{S}$ and an origami $\mathfrak{o}$ which hole ∂ of prefixed diameter d_δ would give the maximum diameter under $\mathfrak{o}$?

3. Which (topological) properties does an origami have as a function?

Example 1. *Suppose you have the configuration, where $\mathcal{S}$ is a square thin piece of paper and ∂ is a equilateral rhombic hole with its angles pointing orthogonally to the surface's sides. Twist the opposite sides of the square at the centres, that should yield a hole shaped like a crystal (triangle + square + triangle, i.e. a hexagon). If the former diameter was d_1 (of the square hole), then after applying such origami and calculating the latter diameter d_2 as described above (orthogonal projection) then the following formula is true: $d_2 = d_1(1 + \cos(\frac{\alpha}{4}))\cos(\theta)$, where α measures how much we twisted the angles of the rhombic*

hole and θ is the angle between the surface and its projection. So, this origami ideally **twices** *the diameter.*